

“Challenge of getting everything to go faster hasn’t eased up a bit”



Teledyne LeCroy provides oscilloscopes, protocol analysers and related test and measurement solutions that enable companies across a wide range of industries to design and test electronic devices of all types. Headquartered in New York, it has sales, service and development subsidiaries in the US and throughout Europe and Asia.

Conrad J. Fernandes, vice president, Asia-Pacific, Teledyne LeCroy, recently spoke to Ashwin Gopinath of EFY about the challenges facing T&M players today, especially when dealing with faster devices.

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What are the major challenges facing T&M players today? How are you trying to overcome them?

One would be the traditional challenge, where we need everything to go faster. That challenge hasn’t eased up a bit. The growth in cloud computing and the need for data rates to go from gigabits to hundreds of gigabits and beyond have only served to create similar challenges all throughout.

Different manufacturers have taken different approaches to tackle this problem and make devices faster. For instance, there is a technology called digital bandwidth interleave. This technology allows any silicon implementation of an amplifier to go twice as fast, by interleaving them. Digital bandwidth interleave involves additional hardware, calibration and digital signal processing at the back end to recover the signal input by the oscilloscope user.

What is the biggest challenge to packing faster signals into smaller time periods?

The world used to be full of square waves. Digital signals were all square waves. These days, nothing is really square. The top section isn’t flat, the bottom isn’t a zero and the edges aren’t perfectly straight. In fact, not only do they have timing issues but they also move around in time.

This jitter becomes very important as you try and pack faster and faster signals into a smaller period of time. Any timing uncertainty, i.e., jitter, creates significant challenges. Hence not only do we need to measure but also characterise jitters and determine where these are coming from. This means that instruments with very, very high timing resolution can help with the above issues.

How is test instruments’ accuracy improving at higher speeds?

We are seeing a couple of interesting things going on. Speed is increasing all the time. Timing resolution has also

gone upwards for a steady while now. Another thing that is becoming more possible is vertical resolution. Our mainstream oscilloscopes are moving up in vertical resolution and what comes with that is increasing vertical accuracy and lower vertical noise.

All these features allow you to get much better representations of signals. They are enabling the emergence of a whole new realm of analysis—stuff that wasn’t previously feasible. For example, oscilloscopes have been able to do fast Fourier transforms (FFTs) for a long time. They’ve been kind of a poor man’s approach to spectral analysis; the spectrum analyser has much greater spectral resolution and accuracy, but as you increase the resolution of the analogue-to-digital converter in a test instrument it becomes much more able to perform some of that analysis.

I don’t think that the day is getting closer when a spectrum analyser will be replaced with an oscilloscope. But I do think that we can give the designers much better tools than what we’ve been able to, as this is the age of increased vertical resolution and reduced noise in the circuit.

How critical is maintaining signal integrity when signals are getting really fast?

As high-speed signals need to be routed into more and more compact and sometimes less friendly environments, signal integrity measurements and techniques become even more important. So we’ve seen things like network analysers being used to ensure the integrity of PCBs. Our signal integrity network analyser has found some really interesting applications there.

As the aforementioned high-speed signals move through the boards, you need to manage the impedance of the boards very carefully. SPARQ, which stands for s-parameters Quick, provides s-parameters for printed circuit boards much faster than what is possible with the use of network analysers.